MUNASHE LANGTON MUKWEYA

Grambling, LA | munashemukweya2022@gmail.com | (318) 690-8862
github.com/Langton49 | linkedin.com/in/munashe-l-mukweya | munashemukweya.com

Research Interests

Development and optimization of statistical and predictive models for extreme climate events in under-studied regions, with a focus on improving model accuracy for regions with sparse observations in Sub-Saharan Africa.

Education

2023–2027 (expected May)

Grambling State University, Grambling, LA
B.S., Computer Science GPA: 3.95/4.00
B.S., Cybersecurity GPA: 3.94/4.00

- *Mathematics & statistics:* Calculus I–III, Linear Algebra, Probability & Statistics I–II, Discrete Structures, Numerical Methods I (in progress)
- *Computing:* Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, Databases & Data Security, Parallel Processing (in progress), Big Data & Cloud Security (in progress)

Research Experience

May–Jul 2026

Data Science Research Intern
Center for Health and Humanitarian Systems (CHHS), Georgia Institute of Technology, Atlanta, GA
Advisors: Arthi Rao (Associate Director, CHHS) and Akane Fujimoto (Research Engineer II)

- Modeled healthcare accessibility as a function of socioeconomic factors and transportation barriers for an NSF-funded project addressing transportation disadvantage in Georgia.
- Cleaned and integrated over 4.2 million records from non-emergency medical transportation (NEMT), American Community Survey (ACS), and CMS provider data into a combined feature set; derived the accessibility target using the two-step floating catchment area (2SFCA) method.
- Fit baseline OLS regression models in Python ($R^2 = 0.28$); after weak fit, moved to a spatial regression model that accounts for neighboring ZCTAs, raising R^2 to 0.31 to support downstream decision-support tools.
- Received the Best in Showcase Poster Award at PIN Emerging Innovators 2026.

Fall 2024 **Undergraduate Researcher, Physics**
Department of Mathematics and Physics, Grambling State University, Grambling, LA
Advisor: Haeyeon Yang (Professor of Physics)

- Contributed to the development of a light-shifting film that converts incoming sunlight to wavelengths solar cells convert more efficiently.
- Measured prototype films with a source meter and solar cell; collected and analyzed data in LabVIEW and Excel.
- Compared current output of film-coated and control cells to quantify the prototype's efficiency gain.

Technical Skills

Programming	Python, C/C++, SQL, C#, JavaScript, TypeScript, HTML/CSS
Data science	NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Jupyter, Google Colab
Tools	Git/GitHub, VS Code, PostgreSQL, Docker, Linux

Honors & Awards

2026	Best in Showcase Poster Award, PIN Emerging Innovators
2023–2026	President's List, Grambling State University (every semester)
2023–2026	Academic Achievement Award, Grambling State University

References

Arthi Rao, Associate Director, CHHS, Georgia Institute of Technology
arthir@gatech.edu

Akane Fujimoto, Research Engineer II, Georgia Institute of Technology
afujimoto@gatech.edu

Vasanth Iyer, Professor, Grambling State University
iyerv@gram.edu